MIL-C-48368A (AR)
10 March 1983
SUPERSEDING
MIL-C-48368 (PA)
24 January 1977

MILITARY SPECIFICATION

CARTRIDGE, HE, 60MM, M720
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements and provides quality assurance provisions for loading, assembling and packing of a high explosive mortar cartridge designated as Cartridge, 60MM, HE, M720.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.1), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-410	- Nondestructive Testing Personnel Qualification and Certification (Eddy Current, Liquid Penetrant, Magnetic Particle, Radiographic and Ultrasonic)
MIL-STD-453	- Inspection, Radiographic
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

9275526	- Cartridge, 60MM, HE, M720
9293287	- Packing and Marking Container, Ammunition, Fiber, PA for 60MM Cartridge, HE, M720 w/Fuze M734
9293439	- Packing and Marking for Container, Ammunition, Metal: PA 70 for 60MM, Cartridge, HE, M720 w/Fuze M734
9293440	- Box, Wirebound, Packing, Ammunition for 60MM Cartridge, HE, M720 w/Fuze M734 Packing and Marking

INSPECTION EQUIPMENT DRAWINGS

7256281	- Alignment Check Plug
7256348	- Alignment Gage
7257962	- Limit Check Gage
9246386	- Inspection Equipment for Measuring Leakage Metal Ammunition Containers

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Clean explosive scrap. Clean scrap (see para. 6.5) produced in regular operations may be remelted and reused provided that it complies with the requirements, except for form, for the grade of high explosive specified. Scrap from floor sweepings, ventilation, and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used.

3.2 Cartridge. The cartridge shall comply with all requirements specified on drawing 9275526, and with all requirements specified in applicable specifications.

3.3 Surface cavitation. Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not exceed in aggregate more than 1/4 of the circumference. The maximum (max.) dimension of any visible cavity in the faced (bottom) surface of the fuze well shall not exceed 1/4 inch in any direction and the total area of all cavities shall not exceed 80 percent of the faced surface area. No individual point of explosive shall extend above the flat surface on which the liner would rest. (Cavities having a max. dimension of 1/16 inch or less shall be disregarded in this computation).

3.4 Foreign matter in projectile cavity. There shall be no foreign matter in the projectile cavity prior to high explosive pouring. Solid explosive shall not be present.

3.5 High explosive on threads. Threads shall be free of high explosive.

3.6 High explosive on projectile body. Exterior surface of the projectile body shall be free of high explosive.

3.7 Ballistic performance (Proving ground). The cartridges shall comply with the following requirements (see para. 4.4.3.2).

3.7.1 Misfire. The cartridges shall not fail to fire from the weapon.

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3.7.2 Duds. The cartridges shall function on impact.

3.7.3 Short rounds. No cartridge shall have a horizontal range less than 80 percent of the average horizontal range when calculated without the short rounds.

3.7.4 Premature burst. There shall be no premature burst in the gun bore or in the flight.

3.7.5 Metal parts separation. There shall be no metal parts separation in the gun bore or in flight.

3.7.6 Range precision. The horizontal range standard deviation shall be no greater than 11 meters when fired at charge zero, no greater than 55 meters when fired at charge two and no greater than 75 meters when fired at charge four.

3.8 Cavitation in high explosive charge (see 4.5.3).

3.8.1 General. Cavities having a maximum (max) projected dimension of 3/32 inch or less shall be disregarded provided that they are separated by 1/2 inch or more of solid explosive.

3.8.2 Maximum projected areas and dimensions of cavities. The aggregate of the projected area of all cavities shall not exceed 1 square inch and no individual cavity shall exceed 1/2 square inch. Cavities located in the area of the charge which extends from the base of the fuze well to 2.5 inches below the base of the fuze well shall not exceed 3/4 inch in any direction and their aggregate projected area shall not exceed 5/8 square inch.

3.8.3 Piping cavity. A piping cavity on the longitudinal axis of the projectile shall be permitted provided that its projected length does not exceed 1 1/2 inches and its width does not exceed 1/4 inch. The projected area of such a cavity shall not be included when determining cavity area in the charge.

3.8.4 Cracked charges and annular rings. Not more than four transverse cracks shall be permitted in any charge. The cracks shall be no wider than 1/32 inch. Longitudinal cracks shall be disregarded. The presence of annular rings with a projected width of 1/16 inch max. and extended from the shell wall 1/4 inch max. into the charge shall not be cause for rejection.

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3.8.5 Base separation. The base separation from the inside of the rear of the shell to the base of the explosive shall be not more than 1/32 inch.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship. Parts and assemblies shall be free of burrs, chips, dirt, grease, rust, corrosion and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specifiction.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance insepection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as required per Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078 and the first article ballistic test shall be rejected if:

a. One or more misfires occur.
b. Five or more duds occur.
c. One or more short rounds occur.
d. One or more premature burst occurs.
e. One or more metal parts separation occurs.
f. Horizontal range standard deviation is exceeded ($70^{o}F$ conditioned samples only).

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48368A (AR)

PARAGRAPH	TITLE: Cartridge, Assembly and Container		SHEET 1 OF 1		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY: N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body Loading Assembly (dwg. 9236378)				
	Examination for defects	25		3.2	4.4.2.1/4.4.2.2
	Cartridge (dwg. 9275526)				
	Examination of defects	25		3.2	4.4.2.4
	Ballistic test (Proving Ground)	135		3.7	4.5.2
	Support Assembly (dwg. 9299245)				
	Examination of defects	15		3.2	4.4.2.5
	Metal Ammunition Container (dwg. 9293439)				
	Examination of defects	5		5.1	4.4.2.8/4.4.2.9
	Air leakage test	5		5.1	4.4.2.9

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, inspection lots of cartridges shall contain:

a. Projectile metal parts assemblies of one lot interfix from one manufacturer.

b. Composition B of one lot interfix from one manufacturer.

c. Fin assemblies of one lot interfix from one manufacturer.

d. Ignition cartridges of one lot interfix from one manufacturer.

e. Propellant charges of one lot interfix from one manufacturer.

f. Obturating rings of one lot interfix from one manufacturer.

g. Fuzes of one lot interfix from one manufacturer.

4.4.2 Examination. See MIL-A-48078 (PA).

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. The contractor may elect to perform 100% inspection with automated equipment or impose mandatory process controls on automated operations in lieu of conducting sampling inspection for defects listed in this specification. In such cases, calibration plans shall be submitted to the Technical Agency (see para. 6.3) for approval for each automatic inspection station and process control plans must be submitted for approval for each of the automated operations. The plans shall identify; (1) the frequency (production interval) for validating equipment functioning, (2) the methods and procedures for validation including designs of defect standards and (3) a description of material control procedures to be imposed on product passing through each automated station during the producton interval between validations. Under these provisions final acceptance will be withheld from material produced during a production interval until satisfactory validation of each of the automated stations so controlled. In the event of a validation failure of any station, the material processed through such station since the last acceptable validation shall be placed in a rejected status pending corrective action.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48368A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Body Loading Assembly, Prior to Loading				9236378
					NEXT HIGHER ASSEMBLY
					9275526
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Foreign object in charge cavity		100%	3.4	Visual
Major	None defined				
Minor					
201	Interior protective coating with bare spot		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48368A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Body Loading Assembly, After Loading		9236378
			NEXT HIGHER ASSEMBLY
			9275526

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	High explosive on projectile body		100%	3.6	Visual
Major					
101	Depth of fuze well, max.		0.40%	3.2	Gage
102	Diameter of fuze well, max.		0.40%	3.2	Gage
Minor					
201	Surface cavitation excessive		0.65%	3.3	Visual
202	High explosive on threads		0.65%	3.5	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48368A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Charge, Propelling		9312698
			NEXT HIGHER ASSEMBLY 9275526

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Container or container seal damaged to the extent that propellant shall escape.	(a)	100%	3.2	Visual (see 4.5.4)
Major					
101	Container or container seal damaged but not to the extent that propellant shall escape.	(a)	100%	3.2	Visual (see 4.5.4)
102	Gross weight of propelling charge, min.	(a)	100%	3.2	Gage
Minor	None defined				

NOTES: (a) If LAP of the propelling charge and cartridge are performed at the same installation, this inspection is not required.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48368A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.4	Cartridge				9275526
					NEXT HIGHER ASSEMBLY N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Minimum distance from top of ignition cartridge assembly to edge of fin, incorrect		100%	3.2	Gage
2	Propellant charge missing or damaged to the extent that propellant shall escape		100%	3.2	Visual (see 4.5.4)
3	Obturating ring missing		100%	3.2	Gage (note a)
Special					
A	Warning label missing or illegible		100%	3.2	Visual
B	Fin blade distorted		100%	3.2	Gage
Major					
101	Obturating ring not flush or below bourrelet diameter		100%	3.2	Gage (Note a)
102	Outside diameter, max.		100%	3.2	Gage
103	Cartridge assembly fails to fully enter profile and alignment gage		100%	3.2	7256348/7256281/7257962
104	Obturating band damaged		0.40%	3.2	Visual
105	Fin assembly torque		0.40%	3.2	Test
106	Fuze torque		0.40%	3.2	Test
107	Ignition cartridge torque		0.40%	3.2	Test
108	Fin assembly not fully seated		0.40%	3.2	Visual
109	Fuze not fully seated		0.40%	3.2	Visual
110	Propellant charge damaged but not to the extent that propellant shall escape		100%	3.2	Visual (see 4.5.4)

NOTES: (a) The inspection for defects can be performed at a station prior to this station when deemed necessary.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.4	Cartridge, cont		9275526
			NEXT HIGHER ASSEMBLY N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Paint damage area in excess of 1/4 square inch		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Pettman cement application improper		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual
	PROVING GROUND TEST				
	Production Ballistic Acceptance	90/48		3.7	4.4.3.2/4.5.2

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48368A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Support Assembly		9299245
			NEXT HIGHER ASSEMBLY 9293287

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Length of spacer		0.65%	5.1	Gage
202	Distance between intermediate leg and forward leg of spacer		0.65%	5.1	Gage
203	Width of smaller opening of spacer and end slots		0.65%	5.1	Gage
204	Height of support		0.65%	5.1	Gage
205	Lettering missing		0.65%	5.1	Visual
206	Support loose		0.65%	5.1	Visual
207	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48368A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Fiber Container (Unsealed)		9293287
			NEXT HIGHER ASSEMBLY
			9293439

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
A	Propelling charge support assembly missing		100%	5.1	Visual
B	Support assembly position improper		100%	5.1	Visual
Major					
101	Tube support missing		0.40%	5.1	Visual
102	Cushion missing		0.40%	5.1	Visual
103	Packing stop missing		0.40%	5.1	Visual
Minor					
201	Round cannot be removed by hand		0.65%	5.1	Visual
202	Glue or asphalt on round		0.65%	5.1	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48368A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Fiber Container (Sealed)				9293287
					NEXT HIGHER ASSEMBLY 9293439
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container damaged or cut through all asphalt impregnated layers		0.40%	5.1	Visual
102	Sealing strip incomplete, badly wrinkled or missing		0.40%	5.1	Visual
103	Metal end loose or distorted		0.40%	5.1	Visual
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	5.3	Visual
202	Contents loose		0.65%	5.1	Manual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Metal Ammunition Container (Unsealed)				9293439
					NEXT HIGHER ASSEMBLY 9293440
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any material containing condensate and/ or surface moisture		0.40%	5.1	Visual
102	Spacer missing		0.40%	5.1	Visual
103	Cushion missing		0.40%	5.1	Visual
104	Incorrect number of assemblies		0.40%	5.1	Visual
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48368A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Metal Ammunition Container (Sealed)				9293439
					NEXT HIGHER ASSEMBLY 9293440
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Air Leakage Test		100%	5.1	4.4.3.1/4.5.1
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	5.3	Visual
202	Contents loose		0.65%	5.1	Manual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48368A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Packing Box (Sealed)				9293440
					NEXT HIGHER ASSEMBLY
					N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Boards split or separated exposing containers		0.65%	5.2	Visual
202	Contents loose		0.65%	5.2	Manual
203	Marking misleading or unidentifiable		0.65%	5.3	Visual

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4.4.3 Testing.

4.4.3.1 Air leakage test (see drawing 9293439) Major defect. The sealed metal ammunition container shall be tested 100 percent for leakage. Any container which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in para. 4.5.1.

4.4.3.2 Proving ground test (see para. 3.7). The cartridges shall be observed for the following characteristics:

a. Misfires, Major defect (see 3.7.1).
b. Duds, Major defect (see 3.7.2).
c. Short rounds, Critical defect (see 3.7.3).
d. Premature burst, Critical defect (see 3.7.4).
e. Metal parts separation, Critical defect (see 3.7.5).
f. Range precision, Major defect (see 3.7.6) ($70^{o}F$ conditioned samples only).

Testing shall be as specified in 4.5.2.

4.4.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, ninety (90) random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. One or more misfires occur
b. Four or more duds occur
c. One or more short rounds occur
d. One or more premature burst occurs
e. One or more metal parts separation occurs
f. Horizontal range standard deviation at charge 0 charge 2 or charge 4 is exceeded.

4.4.3.2.2 After three consecutive lots have complied with the criteria of 4.4.3.2.1, forty-eight random sample cartridges shall be selected from each lot for this test. The lot shall be rejected if:

a. One or more misfires occur
b. Three or more duds occur
c. One or more short rounds occur
d. One or more premature burst occur
e. One or more metal parts separation occur
f. Horizontal range standard deviation at charge 0 or charge 4, if exceeded.

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4.4.3.3 Cavitation (see 3.8) Major defect. The sampling plan and procedure for this test shall be in accordance with MIL-STD-1235 using CSP-2 plan having the following parameters; i value shall be 144, f value shall be 1/24, and the S value shall be 1800. Any cartridge having cavitation in excess of that permitted by the applicable requirement for any segment shall be classed defective. The test shall be performed as specified in 4.5.3.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.2 herein.

4.5 Test methods and procedures.

4.5.1 Air leakage test. The shipping and storage containers shall be tested in accordance with drawing 9246386.

4.5.2 Proving ground test. The test shall be performed at a Government Proving Ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The cartridges for first article testing shall be separated into three equal groups; one group conditioned for 24 hours at plus 145°F ± 2.5°F, one group conditioned for 24 hours at plus 70°F ± 2.5°F, and one group conditioned for 24 hours minus 50°F ± 2.5°F. Each group shall be separated into three equal subgroups; one subgroup tested at charge zero, one subgroup tested at charge two, and one subgroup tested at charge four.

The cartridges for production lot acceptance shall be conditioned at 70°F ± 2.5°F. The sample for the first three production lots (90) shall be separated into three equal groups; one group tested at charge zero, one group tested at charge two, and one group tested at charge four.

After three consecutive lots have complied with the acceptance requirements, the production lot sample (48) shall be separated into two groups; one group tested at charge zero, and one group tested at charge four.

The cartridge shall be fired from a standard 60MM LWCMS mortar at an elevation of forty-five degrees from a hard mount (a solid base test fixture) within two minutes after removal from the conditioning chamber. All fuzes shall be set for impact functioning. The individual velocities, pressures, deflections, and ranges shall be recorded on the firing record.

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Observation shall be made for misfires and duds, and shall be recorded. Should a premature burst or metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The velocity and range standard deviations and arithmetic averages shall be calculated and recorded. Range probable error shall be calculated and recorded. Temperature of the day, weather and the time of firing shall also be recorded.

Hawk radar shall be used to observe for metal parts separation, unobserved short rounds and unobserved duds. In these events, an analysis should be made of the radar data and should be noted on the firing record. This shall include a comparison of times of flight of the unobserved rounds with that of the normal rounds. Prints of the radar traces of these rounds, and prints of the preceding and succeeding rounds, shall be attached to the firing record. Data from ballistically acceptable lots may be discarded.

A smear camera shall be used to photograph all test samples. The camera shall be positioned to pick up each round just outside the second velocity coil. Should an unobserved dud, short round or any other abnormal performance occur, a photo print of these rounds, and prints of all the other rounds in that particular charge zone, shall be attached to the firing record. Film from acceptable lots may be discarded.

4.5.2.1 Test validity. If for any reason the proving ground personnel considers that the test conditions have detrimentally affected the test result, the test results shall be declared invalid and a new test shall be performed on additional sample cartridges.

4.5.3 Determination of cavities. The sample cartridges (HE filled) shall be subjected to radiographic examination. The radiographic beam shall form a 90° angle with the longitudinal axis of the projectile. Examinations of the radiograph shall be made to determine compliance with applicable requirements. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. In addition, the sensitivity of the radiographic technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25, as determined in accordance with 4.5.3.2. In the event an original radiograph leave doubt as to acceptability, the cartridge shall be re-radiographed at an angle 90° from that of the original exposure and in the same plane.

4.5.3.1 Sensitivity check of radiographic technique. The sensitivity of radiographic technique shall be determined by the means of a penetrameter fabricated as specified herein.

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Location of the penetrameter hole shall be 6 3/4 inches measured from base shoulder of projectile body towards fuze nose threads. The projectile, with attached penetrameter, shall be positioned between the radiographic source and the radiographic film with the penetrameter facing the radiographic source. When radiographing a group of projectiles on a rotating ring, the penetrameter, taped to a projectile, may be used to represent the group when checking sensitivity. The sensitivity of radiographic technique shall be considered satisfactory when the image of the penetrameter, including the holes, is sharply defined on the radiograph. The penetrameters shall be made of steel having an approximate specific gravity of 7.8. Penetrameter thickness shall be .020 plus or minus (±) .010 inches. Each penetrameter shall have a circular hole with a diameter twice that of the penetrameter thickness. For further details on construction of the penetrameters, see MIL-STD-453. Frequency of sensitivity check shall be no less than one in each fifty (50) radiographs.

4.5.3.2 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of the maximum equivalent thickness to be radiographed. Film density shall be checked each hour of operation, or fraction thereof, for each film processor. Greater frequency shall be required for any period during which dificulty is encountered in establishing the required densities. To judge densities, it is recommended that standards be used. Comparison of the film with standards may be made visually; however, the use of a densitometer is recommended.

4.5.3.3 Personnel qualification and certification. Personnel performing the radiographic examination in accordance with paragraph 4.5.3 shall meet the qualification and certification requirements defined in MIL-STD-410 as follows:

4.5.3.3.1 Set up and calibration. Level II.

4.5.3.3.2 Operator. Level I.

4.5.3.3.3 Documentation. The contractor will prepare and submitt for approval, in accordance with data item description DI-S-3628 and the DD Form 1423, his detailed procedures for qualification and certification of personnel performing the non-destructive testing procedures required herein.

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4.5.4 Damaged propelling charge. To ascertain that a damaged propelling charge is to be classed as critical or major, the following procedure shall be used:

a. Place the damaged charge on a M27 fin, oriented such that the propellant would be resting against the damaged surface.

b. The fin-charge shall be hand-shaken vigorously three times.

c. Examination shall be made for propellant spillage - if any propellant is observed, the damaged propelling charge shall be classed as a critical defect; if no propellant is observed, the damaged propelling charge shall be classed as a major defect.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified on drawings (dwgs.) 9293287 and 9293439.

5.2 Packing. Packing shall be as specified on dwg. 9293440.

5.3 Marking. Marking shall be as specified on dwgs. 9293287, 9293439 and 9293440

6. NOTES

6.1 Ordering data. See MIL-A-48078 (AR).

6.2 Submission of designs for approval. See para. 6.2.3 of MIL-A-48078 (AR). Submit equipment designs, as required, to Commander, ARRADCOM, ATTN: DRDAR-QAT-I, Dover, New Jersey 07801.

6.3 Submission of calibration and process control plans. Prior to implementation of calibration or process control plans in accordance with para. 4.4.2, plans shall be approved by Commander, ARRADCOM, ATTN: DRDAR-QAT-B, Dover, New Jersey 07801.

6.4 Submission of test data. One copy of proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to each of the following:

Commander
US Army Armament Research and Development Command
ATTN: DRDAR-QAT-B &
DRDAR-LCU-SI
Dover, New Jersey 07801

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6.5 Clean scrap. Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation. Scrap from floor sweepings, ventilation, and suction apparatus (other than that of drilling operations) is not defined as being clean scrap and shall not be used.

6.6 Loading of high explosive. Solid explosive shall not be added to the melted explosive or to the projectile body during pouring or cooling.

6.7 Change from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.8 Data requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
4.5.3.3.3	Nondestructive Inspection Plan	DI-S-3628

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Custodian:
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Preparing Activity:
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(Project 1310-A294)

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